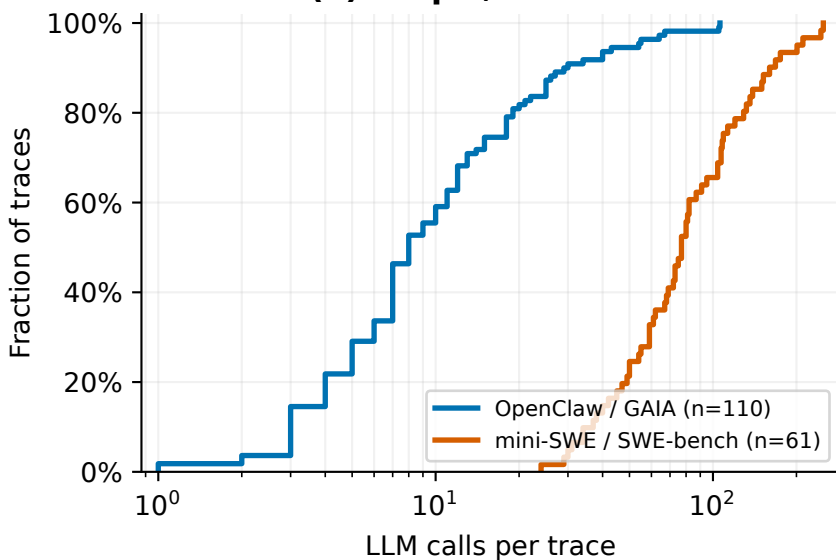
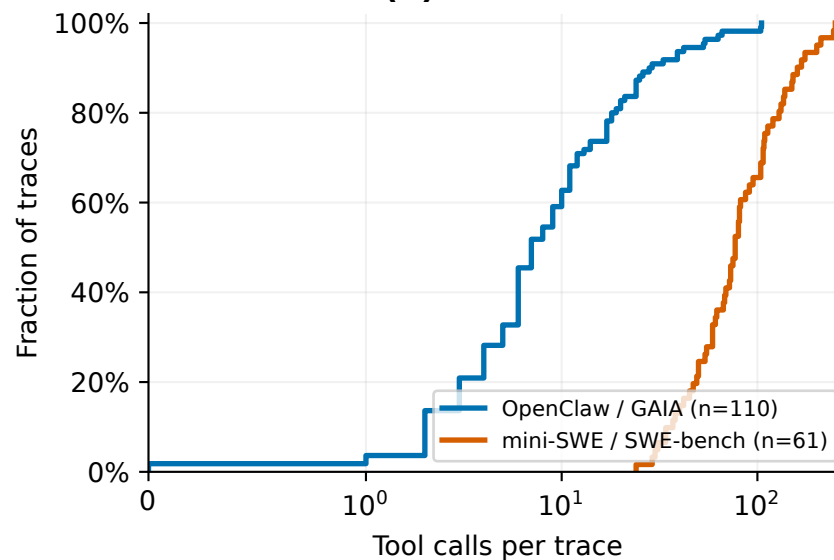


Trace distributions | Qwen scale pool: 110 OpenClaw + 61 mini-SWE

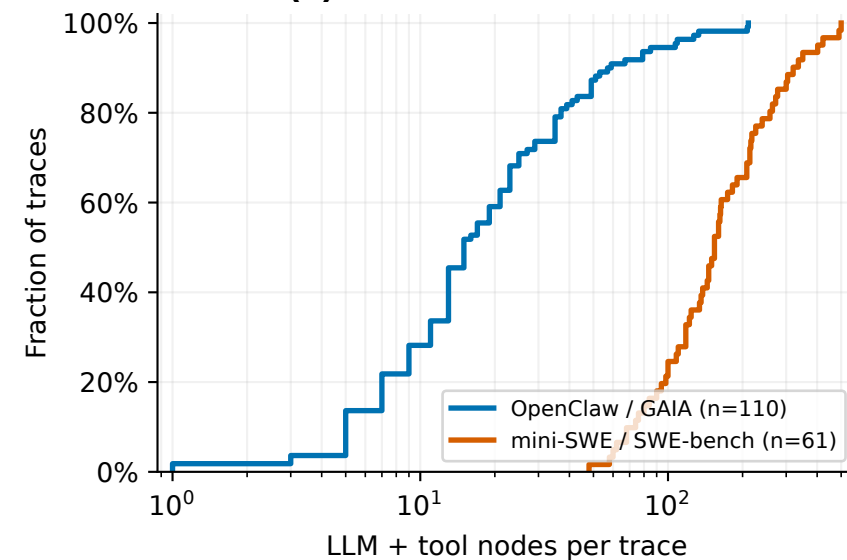
(a) Steps / LLM calls



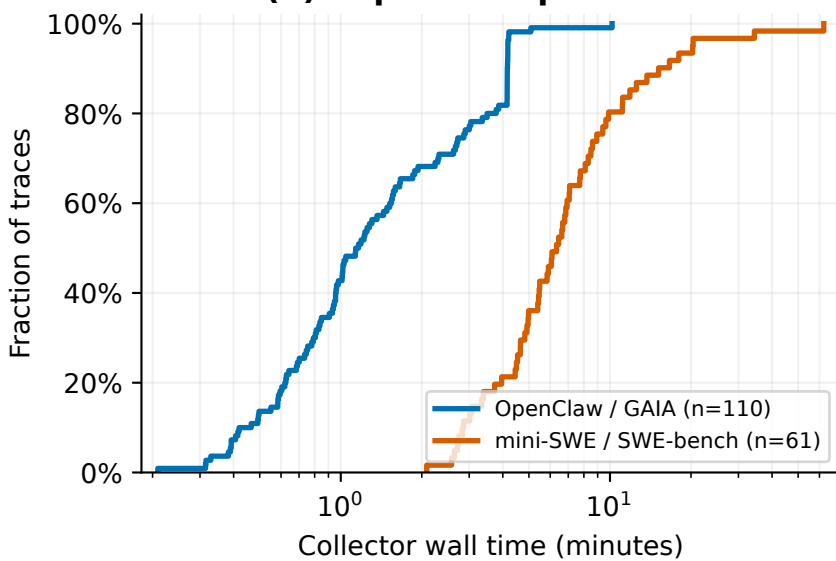
(b) Tool calls



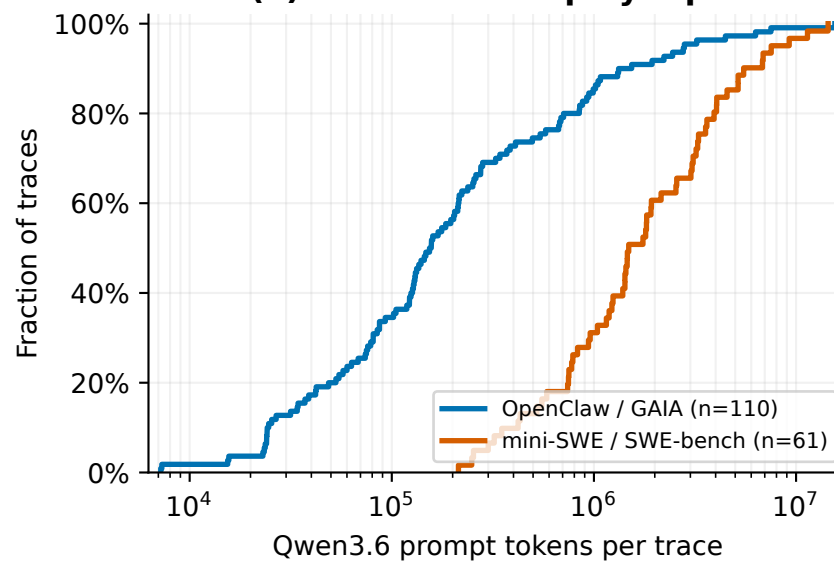
(c) Total recorded nodes



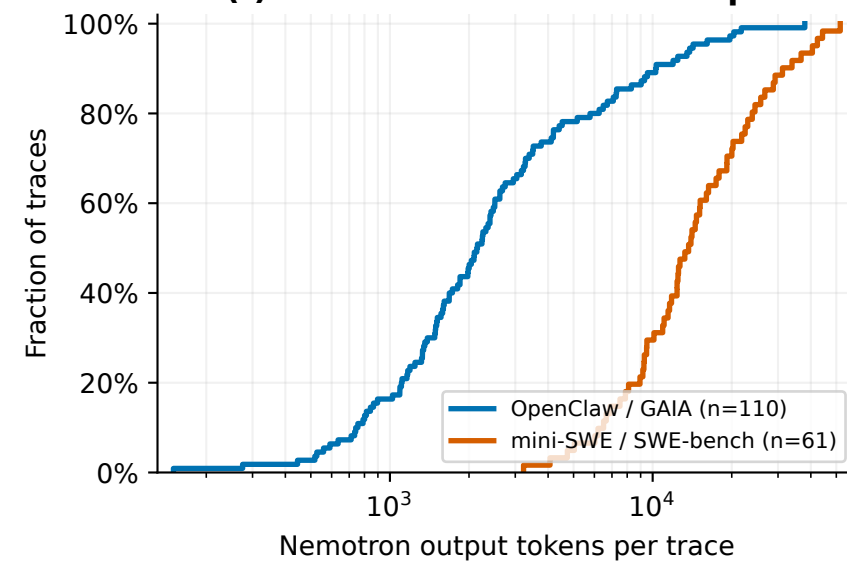
(d) Capture elapsed time



(e) Cumulative replay input



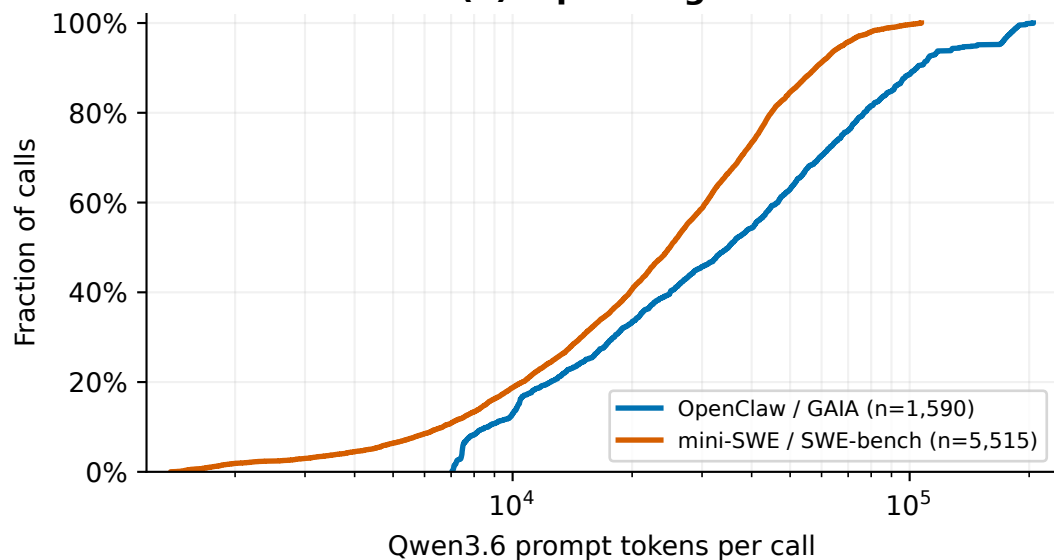
(f) Cumulative recorded output



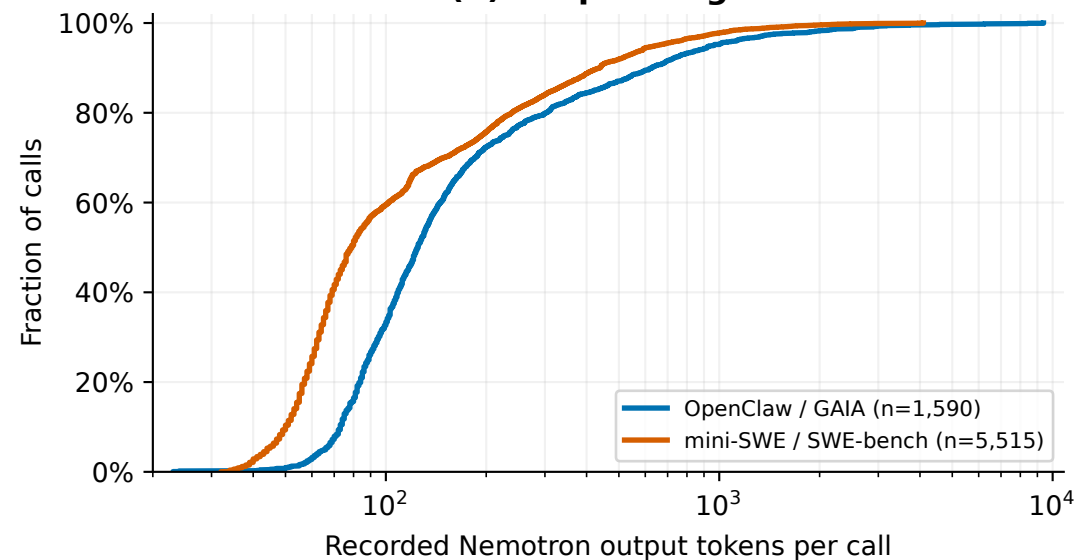
Steps = recorded LLM calls. Input sums count repeated history. Capture time includes endpoint waiting and collector overhead.

Call distributions | calls pooled across selected traces

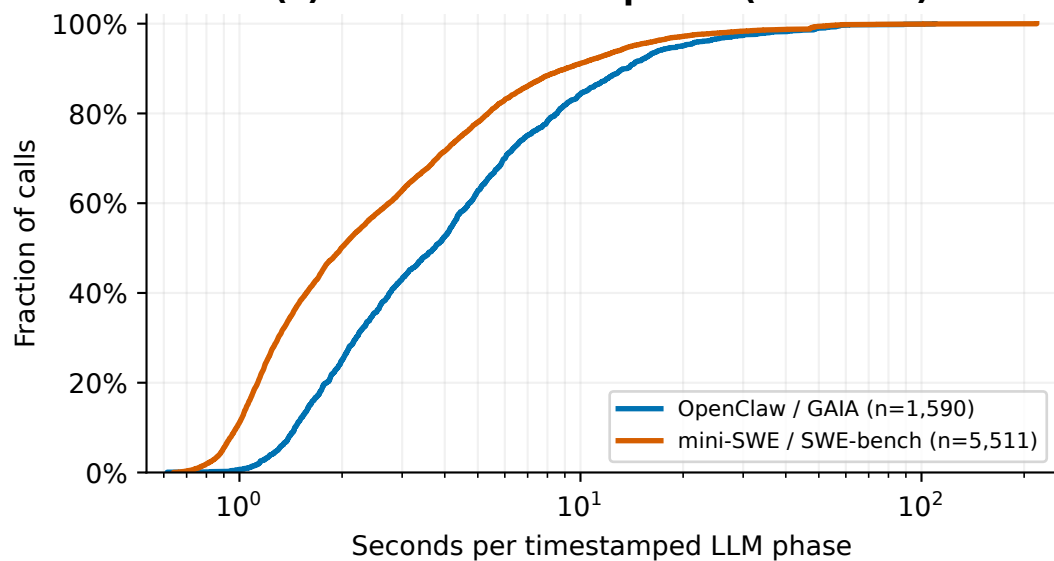
(a) Input length



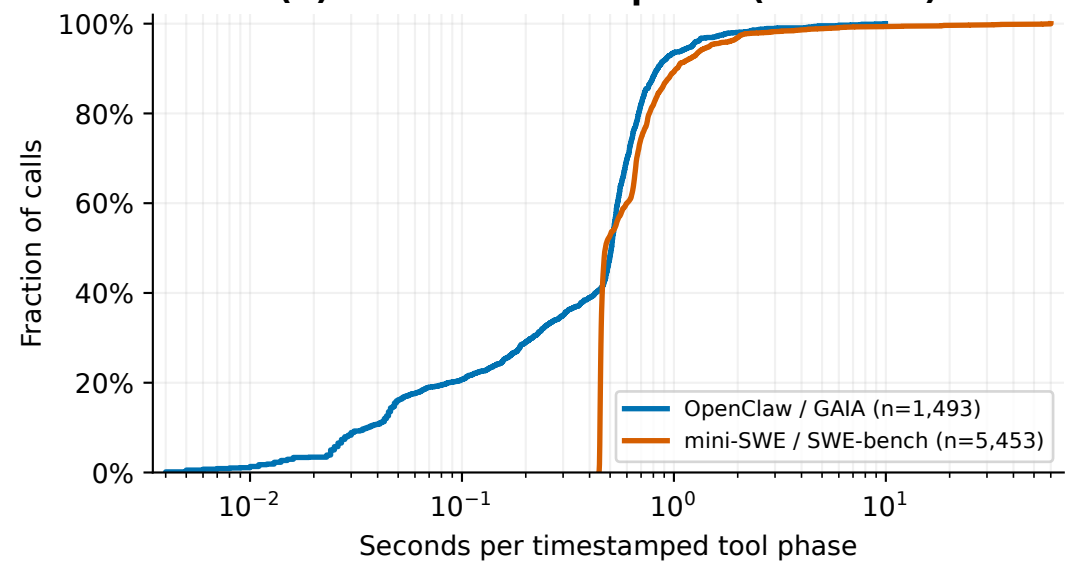
(b) Output length



(c) LLM-associated phase (estimate)



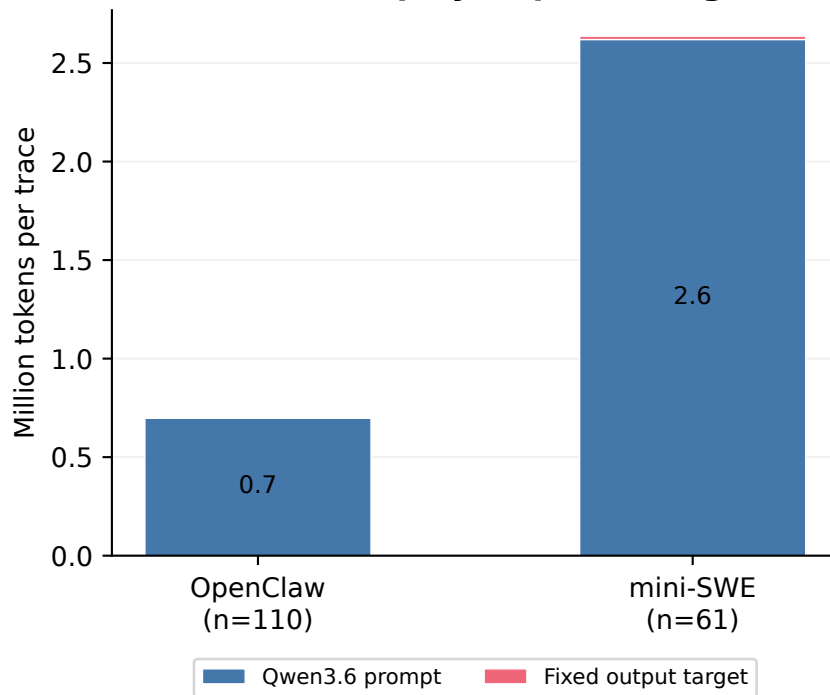
(d) Tool-associated phase (estimate)



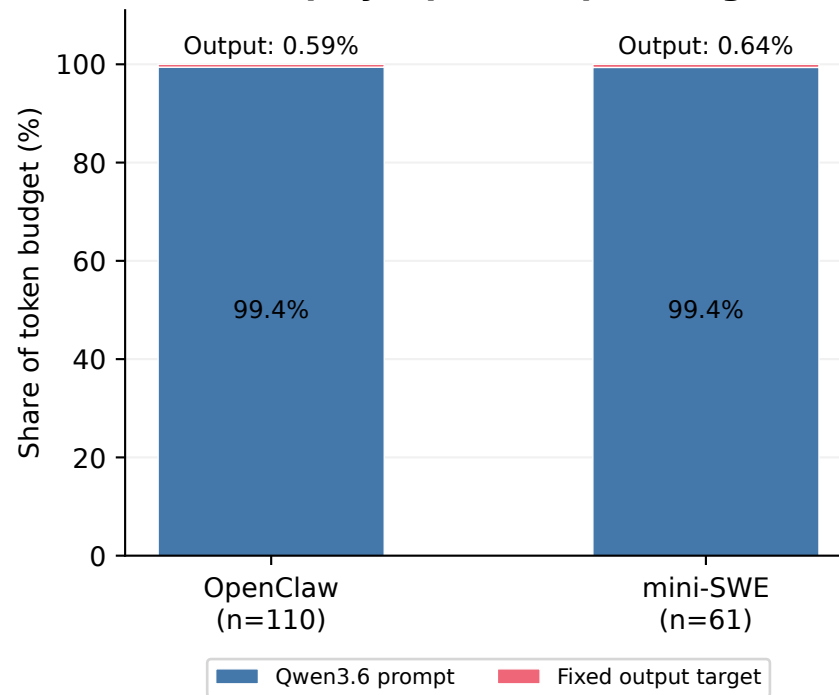
Time panels use event gaps, including orchestration and waiting; missing timestamps are omitted, not filled with zero.
Calls within a trace are correlated; these curves are descriptive, not independent statistical replicates.

Token breakdown | denominators stated separately

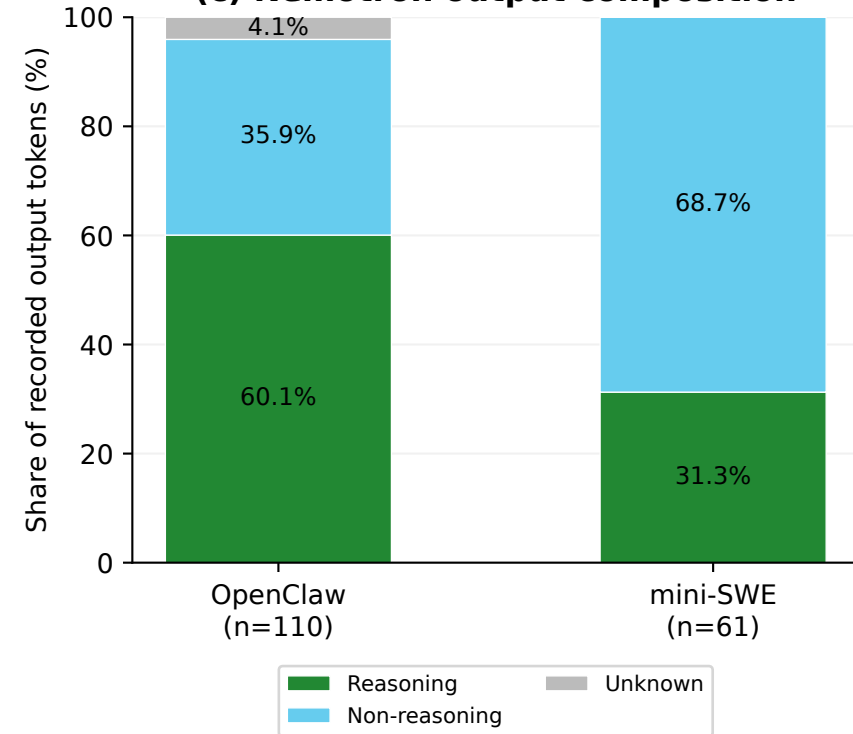
(a) Mean replay request budget



(b) Replay input / output budget

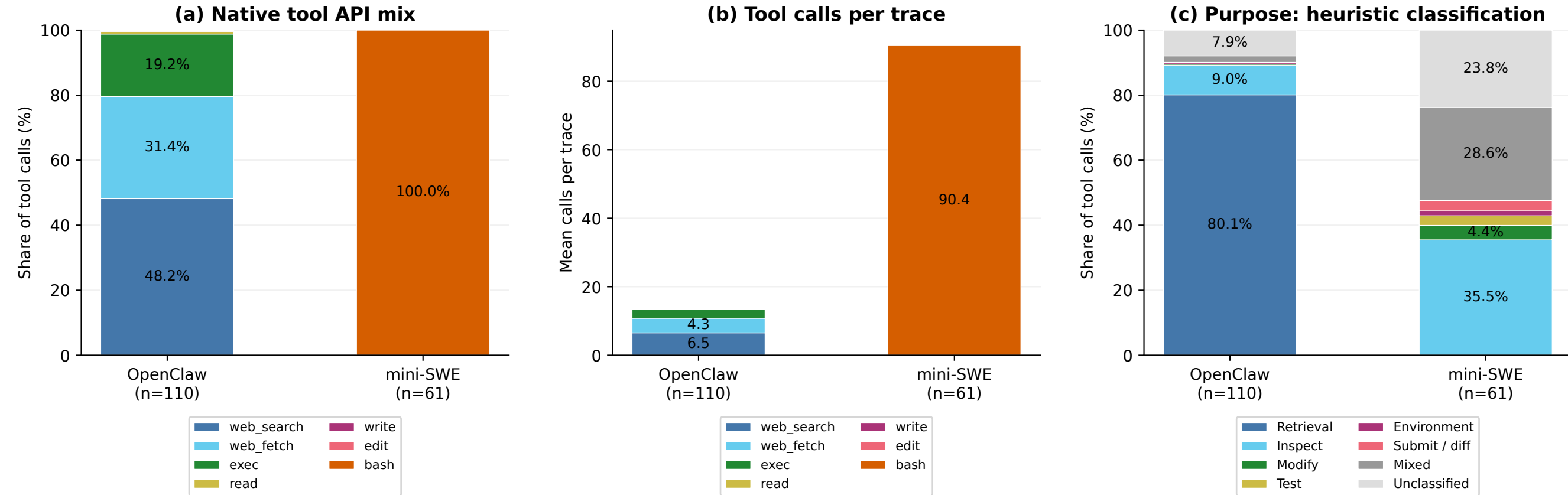


(c) Nemotron output composition



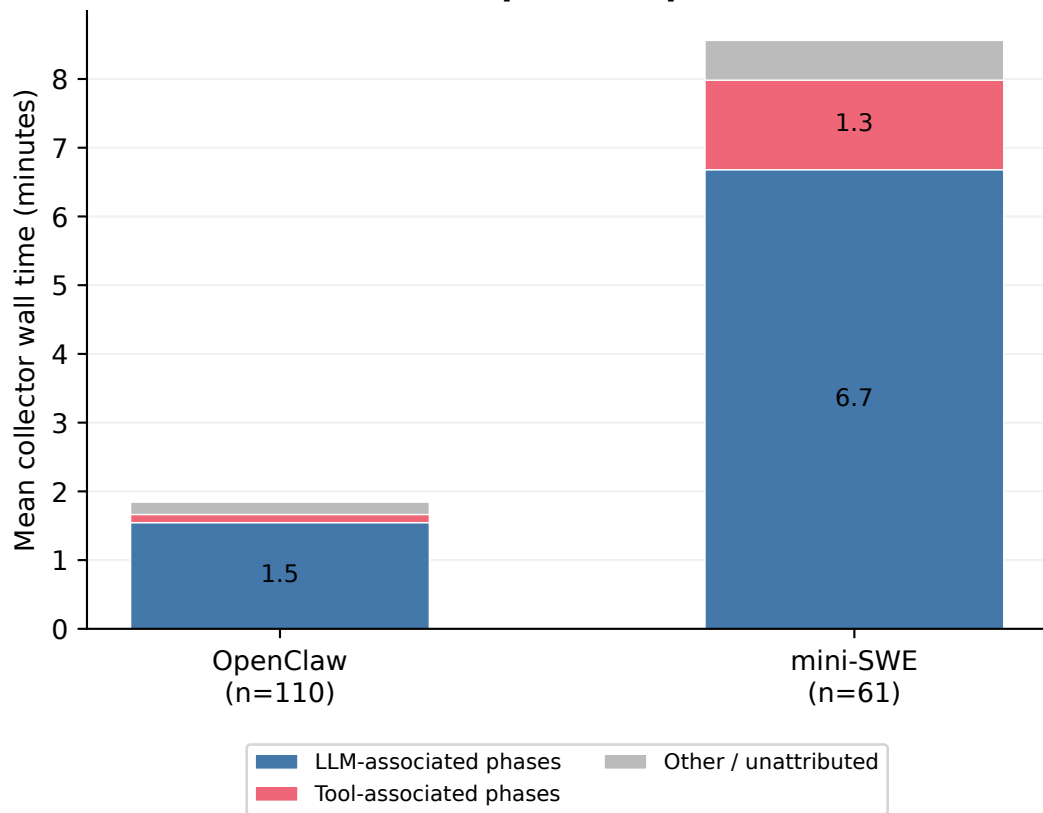
Replay budget combines Qwen prompt counts with fixed recorded output lengths; it is not a Nemotron input/output ratio. Output composition uses native Nemotron usage; non-reasoning includes actions and text. Shares are token weighted.

Tool breakdown | one recorded tool node = one call

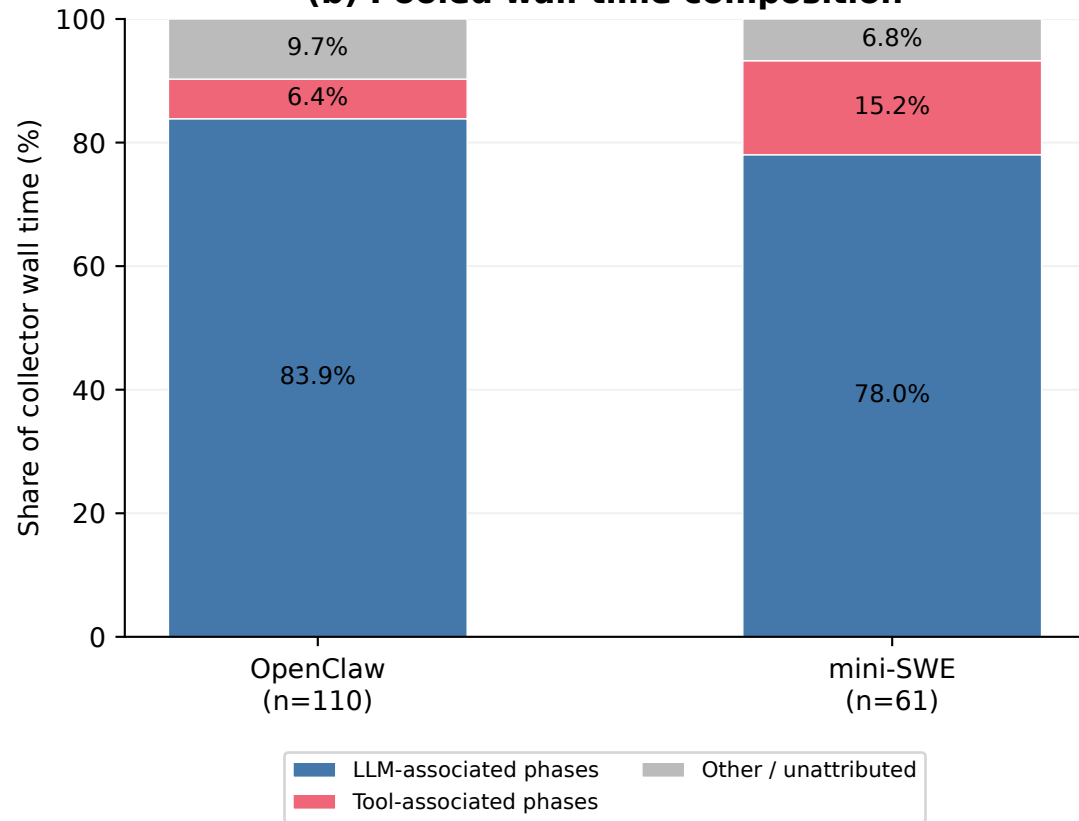


Capture execution-time breakdown | event-gap estimates

(a) Mean capture elapsed time



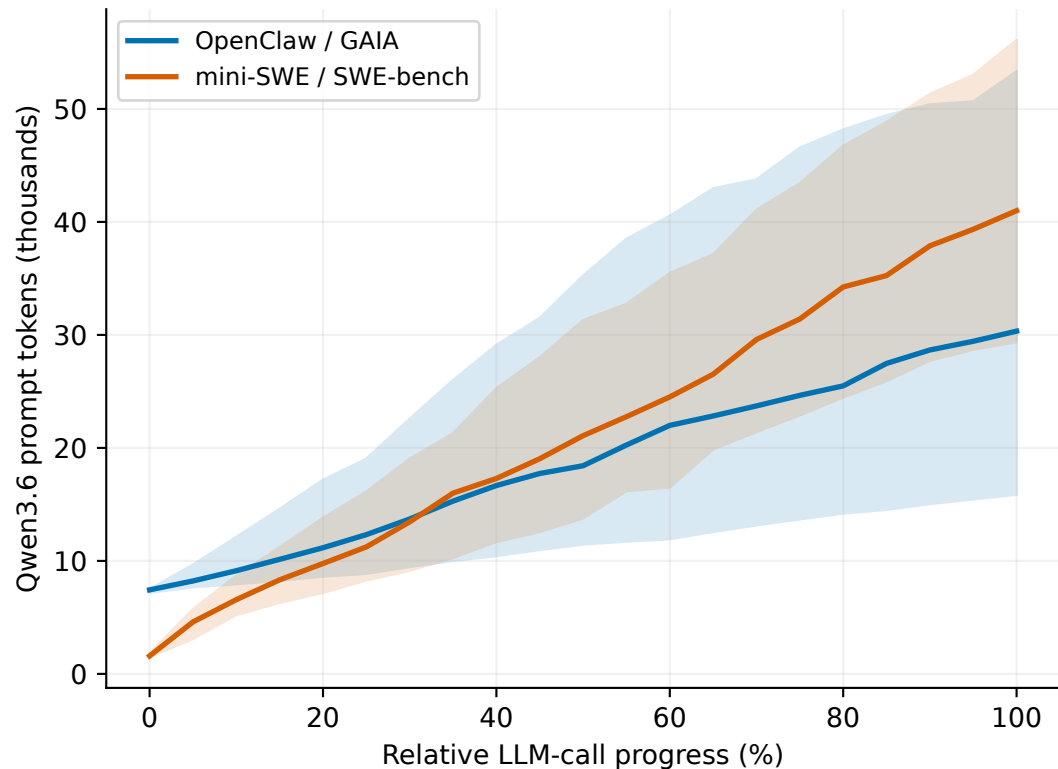
(b) Pooled wall-time composition



LLM phases include endpoint waiting/retries; tool phases include dispatch and logging. These are not GPU compute times. Residual includes setup, cleanup and untimed events. mini-SWE first-call anchor is a provider timestamp (1 s resolution).

Trace progression and capture duration

(a) Context growth: median and IQR



Context curves interpolate within each trace; every trace has equal weight at each progress point.
Capture duration reflects different collection runs and limits, in addition to workload structure.

(b) Steps versus capture elapsed time

